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The apparent polar wander path for the North China Block since the Jurassic

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SUMMARY

A palaeomagnetic study was carried out on sedimentary rocks from Datong, Shanxi Province, and on volcanic rocks from Chifeng, Inner Mongolia and Zhangjiakou, Hebei Province, China. All of these sites belong to the stable part of the North China Block (NCB). After detailed demagnetization experiments, reliable palaeomagnetic directions were obtained for middle Jurassic, late Cretaceous, and late Tertiary. These data define the apparent polar wander path for the NCB since the Jurassic, free from the possible effect of the movement along Tanlu fault since the Mesozoic. Comparisons with reliable palaeomagnetic poles obtained in Siberia and South China indicate that the middle Jurassic poles for North and South China are almost the same, while the Siberian pole is displaced from them to the west by about $18° \pm 13°$. On the other hand, the Cretaceous poles for the three areas are not significantly different from each other. Consequently, it is suggested that the amalgamation of North China and South China was almost complete in the middle Jurassic, and that their accretion to Siberia continued until the late Cretaceous.

Key words: apparent polar wander path, Jurassic to Tertiary, North China, palaeomagnetism.

1 INTRODUCTION

It is now widely believed that China is composed of several blocks which were brought together since the Palaeozoic time and amalgamated one after another (e.g., McElhinny *et al.* 1981; Zhao *et al.* 1990; see Fig. 1). Such tectonic movements can most conveniently be studied using palaeomagnetic methods. Although the increase of palaeomagnetic data in the recent years from China is remarkable, reliable data are still not so numerous as to enable a detailed reconstruction of the formative process of NE Asia. One of the reasons for this is the frequent occurrence of strong overprinting on the natural remanent magnetization (NRM) in many of the Chinese rocks. The other reason, especially for younger ages, is the fact that materials suitable for palaeomagnetic study are not abundant for the North China Block (NCB) for Cretaceous and Tertiary, as most of the NCB continued to be above sea level for that period (Institute of Geology, Chinese Academy of Geological Sciences 1985).

On the other hand, important tectonic developments occurred in geologically young ages in southern and eastern Asia. Opening of many of the marginal basins in the western Pacific took place in middle to late Tertiary. Moreover, the biggest tectonic event in Asia, the collision of India and Eurasia and its effects range from about 45 Ma to the present (Molnar & Tapponnier 1975).

Among the other important tectonic events in eastern Asia is the opening of the Japan Sea which is presumed to have occurred in Miocene. Based on the rapid clockwise rotation of SW Japan revealed by palaeomagnetism (Otofuji & Matsuda 1983), Otofuji, Matsuda & Nohda (1985) proposed that the Japan Sea opened within about 1 Myr at about 15 Ma. However, the problem seems to be still far from being solved: different opinions were expressed by various authors about the mode of the opening (e.g., Lallemand & Jolivet 1986; Celaya & McCabe 1987). The recent result from Ocean Drilling Project Leg 127 seems to contradict a very rapid opening suggested by palaeomagnetism (Shipboard Party 1989, personal communication). A major source of the variety in interpreting the opening of the Japan Sea comes from the fact that while we have a

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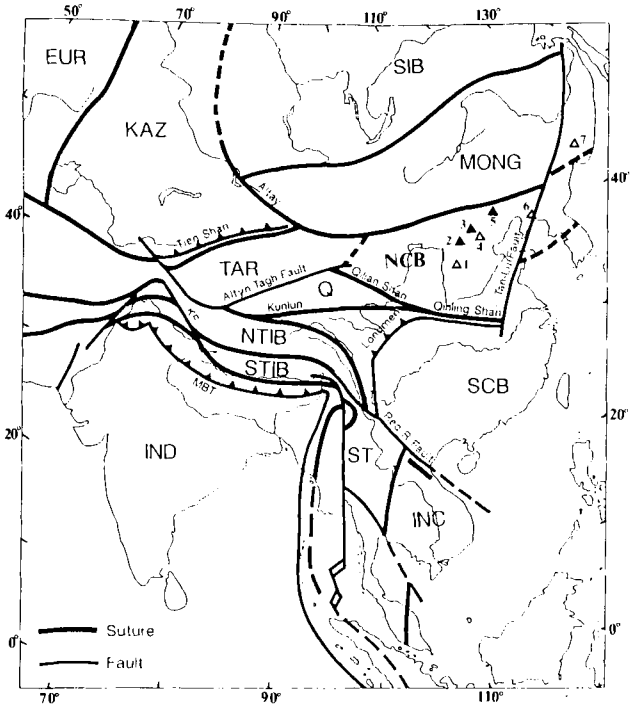


Figure 1. Generalized tectonic map of China (base map by Enkin *et al.* 1990) with sampling localities for the Japan–China cooperative palaeomagnetic study shown by numbers and triangles: solid triangles for the regions discussed in the present paper; (1) Jingle, Shanxi, (2) Datong, Shanxi, (3) Zhangjiakou, Hebei, (4) Beijing, (5) Chifeng, Innermongolia, (6) Benxi, Liaoning, and (7) Qitaihe, Heilonjiang.

comprehensive data set on Japanese islands, we lack them in the Japan Sea or on the continent bordering the Japan Sea, e.g. Sikhote Alin, NE China, and easternmost Siberia. The establishment of the apparent polar wander path (APWP) for North China for geologically young ages is in this respect very important.

Since 1986, a cooperation program started between palaeomagnetists in Japan and China to obtain a reliable APWP for the NCB. Sampling was carried out in Shanxi, Heilonjiang, Liaoning, and Hebei Provinces, Beijing City, and Inner Mongolia in 1987 and 1988 (Fig. 1). In this paper, we present palaeomagnetic results obtained from the Shanxi, Hebei and Inner Mongolia rocks, and discuss their implications to the tectonic evolution of eastern Asia. Palaeomagnetic results from the east of Tanlu Fault differ significantly from those obtained from the west of Tanlu Fault, and they will be reported separately (Uchimura *et al.*, in preparation).

2 GEOLOGICAL SETTING AND SAMPLING

In each of the sampling sites listed below, about 10 samples were obtained by portable engine drills as cores of 25 mm diameter and 50–100 mm length. Orientation of cores was determined by a sun compass when the sun could be sighted, and by magnetic compass otherwise. Local magnetic declination was corrected by using the values in the large-scale maps when they are available, and by IGRF values elsewhere. In the Quaternary volcanic field to the east of Datong, we encountered strong magnetic anomalies which caused appreciable deflection of the compass needle. In every sampling site reported in the present study, however, the intensity of the NRM was not so strong as to cause a noticeable magnetic deflection. So the orientation errors may not exceed 5° even for the ones determined by magnetic compasses. Specimens about 22 mm long were cut from the cores and used in palaeomagnetic measurements.

2.1 Datong area, Shanxi Province

The Shanxi platform is in the centre of the North China Block (NCB) and is one of the stable parts of that block. The sedimentary layers with Palaeozoic and Mesozoic ages are widely distributed, and they are gently folded due to later tectonic movements. The Datong area is located in the north-central part of the Shanxi platform (Fig. 1). The metamorphic rocks of Sanggan formation (Archaean) are covered by Phanerozoic sedimentary layers with ages between middle Cambrian to middle Jurassic. A small patch of Cretaceous limestones is exposed in the western Datong area. In the NE part, Quaternary monogenic volcanoes distribute near Xiping, to the east of Datong city (Hu & Liu 1987).

In the Datong area, upper Permian and Triassic strata, which are widely distributed in Shanxi Province, are absent and the lower Jurassic Yongdingzhuang formation unconformably overlies the Palaeozoic strata. This unconformity is interpreted as a consequence of regional folding and faulting in eastern half of the NCB due to collision–extrusion tectonic movement between the NCB and the South China Block (SCB) (Kimura, Takahashi & Kono 1990). The area to the east of Datong (Hengshan and Wutaishan) uplifted, while the area west of Datong subsided. Jurassic to Cretaceous sedimentation in Shanxi Province was controlled by this tectonic movement. To the west of Datong City, we collected some samples from steeply inclined (>60°) Mesozoic layers in the hope of obtaining positive fold test. In the Quaternary period, extension tectonics prevailed in the east of Datong area, and numerous volcanoes were formed by this activity.

Palaeomagnetic samples were collected near Zuoyun quarry to the west of Datong city and from a road cut near Yungan cave (famous for Buddhist sculptures). In Zuoyun quarry, very flat limestone about 30 m thick (Zuoyun formation) overlies Cretaceous–Jurassic andesites (Baiqi formation) and middle Jurassic sandstones and mudstones of the Yungan formation. The limestones are apparently distributed in a limited area. A standard geological reference (Li *et al.* 1986) only mentions sandstones and mudstones in this region. Late Cretaceous fossils are reported from the Zuoyun formation (Li *et al.* 1986). Core samples were obtained at four sites from the Zuoyun formation, one site from the Baiqi formation, and two sites from the Yungan formation.

Sedimentary formations of Jurassic age (Datong formation and Yungan formation of middle Jurassic and Yongdingzhuang formation of early Jurassic) are well exposed in the Datong area usually with a very gentle dip. Besides the sites at Zuoyun quarry, these rocks were sampled at four sites in Qingciyao, 10 km west of Datong city, two sites from a road cut near the famous Yungan

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Table 1a. Field data of samples from the Datong area.

Site ID	Site		Formation	Age	Rock Type	Bedding	
	lat.	long.				Dip	Strike
DZ01-04	40.12	112.93	Zuoyun	K2	Limestone	3	312
DZ05	40.12	112.93	Baiqi	J3/K1	Basalt	3	312
DZ06	40.12	112.93	Yungang	J2	Siltstone	0	0
DZ07	40.12	112.93	Yungang	J2	Mudstone	0	0
DY07	40.12	113.15	Yungang	J2	Sandstone	0	0
DY08	40.12	113.17	Yungang	J2	Sandstone	16	329
DY02-03	40.07	113.17	Datong	J2	Sandstone	69	178
DY04	40.07	113.17	Datong	J2	Sandstone	71	175
DY05	40.11	113.02	Datong	J2	Sandstone	12	322
DY06	40.11	113.02	Datong	J2	Sandstone	8	315
DY01	40.07	113.17	Yongdingzhuang	J1	Sandstone	75	159

cave, and two sites at a highway near Jinhuagong. Details of all the sampling sites are summarized in Table 1(a).

2.2 Neogene basalts in Chifeng, Inner Mongolia and Zhangjiakou, Hebei Province

Tertiary flood basalts are widely distributed in the north of Hebei Province and southern part of Inner Monoglia. They are exposed in a 200 km wide belt along Jiening–Zhangjiakou–Chifeng. Based on about 20 K–Ar ages obtained for the basalt samples from a drill well in Zhangjiakou area, Chen, Liu & Sun (1985) suggested that the basalts of the whole area belong to the Neogene age. According to Zhou *et al.* (1988) the accumulation of these basalts occurred in three stages. The first eruption period was Miocene with isotopic ages between 25.3 and 19.4 Ma. Alkali basalts and tholeiite belonging to this period are found near Zhangjiakou, Weichang, and Pingquan. The second period was early to late Miocene with ages between 18.7 and 7.5 Ma, when tholeiites and alkali basalts alternated. These rocks are found mainly in the Zhangjiakou area. In the youngest eruption period (5.4–0.2 Ma), tholeiites were accumulated in Sanyi near Zhangjiakou which are dated at about 4.3 Ma.

These basaltic rocks typically form platforms a few kilometres across, with altitudes as high as 400 m. In Chifeng area, basaltic rocks are exposed with well-developed columnar joints. Unfortunately, no isotopic ages are available for these rocks, but Chen *et al.* (1985) suggested that they may belong to the second eruption period. Sun (1987) reported some stratigraphic evidence of the Quaternary age of the basalts in Chifeng area. In a section in Hexigten Qi about 150 km northwest of Chifeng city, middle Pleistocene clays were found below the basalt rocks at a depth of 15–20 cm. In the other section near Chifeng city, Malan loess of late Pleistocene was located under the basalt rocks. Sun (1987) suggested that though the Neogene basalt rocks were dominant, the Quaternary basalt rocks were also distributed in Chifeng area.

Samples were collected at 41 sites in the Chifeng area, and at 10 sites in the Zhangjiakou area. Zhao *et al.* (1990) also sampled the same basalt series at 13 sites in the Jiening and Chifeng areas. In basaltic platforms, most of the samples were collected from quarries where vertical columns are well developed. Care was taken to sample several sites for each platform. Where vertical as well as slanted columnar joints were observed, samples were obtained from both places in the hope of performing a kind of fold test to see if the slanting of the columns occurred later than the eruption. Detailed description of the sampling sites is summarized in Table 1(b and c).

Table 1b. Field data of the Neogene basalt samples from Chifeng, Inner Mongolia area.

Site ID	Site		Bedding	
	lat.	long.	Dip	Strike
CP01-03,				
06-07	42.00	119.30	0	0
CP04	42.00	119.30	41	170
CP05	42.00	119.30	51	60
CP08	42.00	119.30	42	26
CP09	42.00	119.30	41	26
CP10	42.00	119.30	32	65
CX02-07	42.33	118.72	0	0
CJ01-02	42.40	118.60	0	0
CSZ01-02	42.37	118.62	0	0
CG03	42.54	118.87	0	0
CB01- 03,				
05-06	42.32	119.13	0	0
CB04	42.31	119.13	66	245
CW01-03	42.35	118.75	0	0
CH01	42.25	119.02	29	30
CMJ01-02	42.30	118.93	0	0
CHD01-03	42.32	118.90	0	0
CM01	42.13	119.23	20	55
CM02	42.13	119.23	46	10
CM03	42.13	119.23	34	25
CM04	42.13	119.23	36	12
CS01	42.20	119.10	90	210
CS02	42.20	119.10	54	205

Table 1c. Field data of the Hannouba Fm. Basalt samples from Zhangjiakou, Hebei province.

Site ID	Site		Bedding	
	lat.	long.	Dip	Strike
ZY01	41.14	114.61	0	0
ZY02	41.16	114.65	0	0
ZG01	41.04	114.72	0	0
ZL01-03	41.00	114.72	0	0
ZL04	41.00	114.72	28	84
ZL05	41.00	114.72	35	310
ZL06-07	41.00	114.72	0	0

3 PALAEOMAGNETISM

All the samples were measured at Tokyo Institute of Technology using a 2G superconducting magnetometer and a Schonstedt spinner magnetometer. From each site, five to 12 samples were subjected to either alternating field (AF) or thermal stepwise demagnetization. However, stepwise AF demagnetization was not effective for the sedimentary samples and there were hardly any changes in the NRM directions and intensities even at the maximum demagnetizing field (40 mT). Accordingly, the results reported for them are mainly based on stepwise thermal demagnetization, which was carried out using temperature steps between 50° and 100 °C in most cases, but 10° to 20 °C when necessary, and 10–16 steps. For volcanic rocks, both stepwise thermal

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and stepwise AF methods were effective. The directions of primary remanences for the volcanic rocks were derived from the results of both methods. The direction of primary magnetization was determined from the linear portion of orthogonal plot by least-squares fitting of a straight line. The tilt of strata was corrected by using the bedding angles listed in Table 1.

3.1 Jurassic rocks (Yungan, Datong, and Yongdingzhuang formations)

The Yongdingzhuang formation was sampled at one site (DY01), but its remanence was unstable to thermal demagnetization and no reliable results were obtained.

In the Datong formation, three sites (DY02, DY03, and

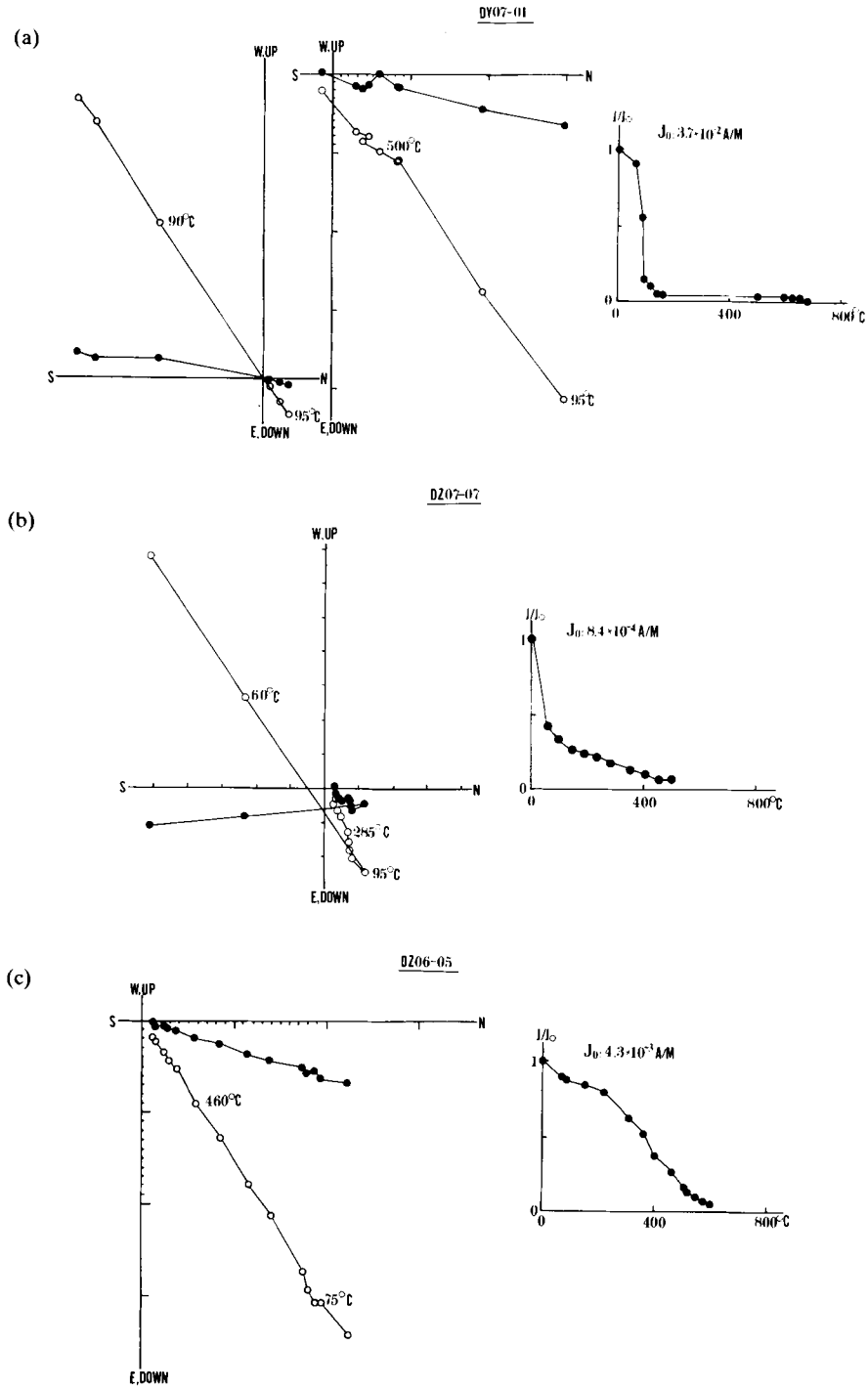


Figure 2. Examples of orthogonal projections of the magnetization vectors in Jurassic rocks in the Datong area during the course of thermal demagnetization. Insets show the decay of intensity of remanence by demagnetization. In the orthogonal projections (this figure and the following), solid (open) circles are those projected on the horizontal (vertical) plane in geographical coordinates. The NRM intensities are given in the figures. (a) DY07–1 from sandstone, (b) DZ07–7 from mudstone, (c) DZ06–5 from siltstone of the Yungan formation. The unit on the vector plot scales is 10^{-4} A m^{-1}.

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DY04) showed a single component of magnetization very close to the present axial dipole direction. Upon thermal demagnetization, the NRM intensity decreased rapidly and less than 10 per cent of the original remanence remained by 250 °C (DY04) or 350 °C (DY02, DY03). These sites have large bedding angles (dips of about 70°). After tilt correction, the remanence direction becomes very shallow and westerly. Such a shallow inclination is not consistent with Jurassic or later palaeomagnetic directions from this study as well as from other authors. Undoubtedly, these magnetizations were acquired after the tilting occurred.

The NRMs of DY05 and DY06 are very stable and contain only one component which persists up to 400 °C. The *in situ* direction of DY05 is not statistically different from the present axial dipole direction and that of DY06 is only marginally different (3.7° against the α_{95} of 3.5°). As the bedding angles are similar for both layers, tilt correction only moves the remanence directions by similar amounts. Although the tilt corrected directions of the NRM are reasonable for the Jurassic, it cannot be proved that they are primary. Therefore, we reject these two data as there is some doubt about remagnetization.

Among the sites belonging to Yungan formation, DY08 was unstable and no significant results were obtained from it. The NRM directions are almost in the present field direction. Upon stepwise thermal demagnetization, the direction of remanences moved erratically and the stable direction cannot be identified. For other sites (DY07, DZ06, DZ07), stable NRM directions were identified by stepwise demagnetization (Fig. 2a, b and c). The NRMs of DY07 and DZ07 showed a very queer phenomenon. The original remanence points south and upward, a very typical 'reversed' direction. Upon demagnetization, however, it reverts to normal direction by heating only to 100 °C (Fig. 2a and b). By demagnetization at 150 °C, the intensity of remanence became less than one tenth of the original NRM. After that, the NRM stays in the north and downward direction to 650 °C without further changes. Remarkably similar behaviour was observed for sandstone (DY07) and mudstone (DZ07) samples. The low-temperature component which can be demagnetized at 100 °C seems to be carried out by goethite. The weak but very stable high-temperature components must be carried by haematite, and give a direction which is significantly different from the present-day field direction. The sandstone layer DZ06, which overlies the mudstone DZ07, has a very stable remanence persisting up to 600 °C, with a direction similar to the high-temperature components of DY07 and DZ07. Interestingly enough, very similar behaviour of the NRM was found by Yang Zhengyu from the Jurassic Anding formation in Shaanxi Province about 500 km away from Datong (Randy Enkin & Yang Zhenyu, personal communication 1990).

Following the demagnetization experiments, we conlcude that primary direction was obtained from at least three sites (DY07, DZ06, DZ07) (Fig. 5a). The mean direction of magnetization for the three sites is $I = 58.7°$, $D = 17.9°$ ($\alpha_{95} = 5.9°$ $k = 443$). Results of palaeomagnetic measurements of Jurassic rocks are summarized in Table 2. This Jurassic direction is not substantiated by the fold test or reversal test, because all the rocks have normal polarity and the accepted layers are nearly horizontal. As the NRMs in

Table 2. List of palaeomagnetic results from the North China Block.

Site	Age	polarity ratio R/N	N/N0		Directions: before tilt correction I(°)	D(°)	α_{95}(°)	k	Directions: after tilt correction I(°)	D(°)	α_{95}(°)	k		VGPs: *Lat* (°N)	*Lon* (°E)	*dp* (°)	*dm* (°)
DY01	J_1	0/10	10/10	NRM	37.5	312.3	56.4	1.7	-10.7	295.2	56.4	1.7	G	45.4	12.4	39.1	66.4
DY08	J_2	0/6	6/6	NRM	54.5	2.3	10.6	41.1	60.5	338.6	10.6	41.1	G	84.6	273.3	10.5	14.9
DY02	J_2	0/6	6/6		55.5	354.9	7.0	92.5	15.5	303.9	7.0	92.4	G	84.3	339.4	7.1	10.0
DY03	J_2	0/6	5/6		53.7	8.7	3.9	387.7	23.0	307.2	3.9	387.9	G	81.0	240.8	3.8	5.4
DY04	J_2	0/5	4/5		53.0	0.9	8.9	107.6	18.6	304.2	8.9	107.6	G	83.4	286.6	8.5	12.3
DY05	J_2	0/7	5/7		53.6	6.9	3.1	596.6	44.4	15.9	3.1	597.2	G	81.9	248.6	3.0	4.4
DY06	J_2	0/6	6/6		61.2	6.5	3.5	365.3	54.3	13.0	3.5	365.3	G	84.6	176.6	4.1	5.4
Mean (DY02-06)			5/5		55.5	3.5	4.4	302.0	35.4	325.4	35.2	5.7	G	85.2	256.8	A_{95} =5.5	

Site	Age	polarity ratio R/N	N/N0		Directions: before tilt correction I(°)	D(°)	α_{95}(°)	k	Directions: after tilt correction I(°)	D(°)	α_{95}(°)	k		VGPs: *Lat* (°N)	*Lon* (°E)	*dp* (°)	*dm* (°)
DZ06	J_2	0/6	5/6		60.0	11.5	8.9	75.0	60.0	11.5	8.9	75.0		81.2	194.1	10.2	13.4
DZ07	J_2	0/7	6/7		59.1	24.9	4.5	226.3	59.1	24.9	4.5	226.3		71.0	195.6	5.0	6.7
DY07	J_2	8/0	8/8	L	-57.7	188.6	2.3	596.1	-57.7	188.6	2.3	596.1		83.1	215.3	2.6	3.6
		0/8	8/8	H	56.7	17.3	2.5	489.9	56.7	17.3	2.5	489.9		76.2	209.4	2.6	3.6
Mean (DZ06-07, DY07H)			3/3		58.7	17.9	5.9	443.1	58.7	17.9	5.9	443.1		76.2	199.9	A_{95} =8.3	
DZ05	J_1/K_1	0/6	6/6		75.2	28.4	3.5	377.4					G	62.1	141.3	5.8	6.3
DZ01	K_2	9/3	11/12		-68.6	184.3	4.3	113.7	-66.1	188.5	4.3	113.5		-79.7	326.2	5.8	7.1
DZ02	K_2	0/5	5/5		62.6	4.9	7.7	100.9	60.2	8.1	7.6	100.9		83.8	191.1	8.8	11.6
DZ03	K_2	0/7	7/7		68.5	15.0	3.0	403.7	65.8	18.0	3.0	408.2		74.8	161.8	4.0	4.9
DZ04	K_2	0/6	6/6		64.1	12.5	5.5	148.6	61.4	15.2	5.5	148.6		78.3	186.0	6.5	8.5
Mean (DZ01-04)			4/4		66.0	9.1	4.3	467.5	63.4	12.4	4.2	473.7		79.6	170.1	A_{95} =5.8	

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Table 2. (*continued*)

Site	Age	polarity ratio R/N	N/N0		Directions: before tilt correction I(°)	D(°)	α_{95}(°)	k	after tilt correction I(°)	D(°)	α_{95}(°)	k		VGPs: *Lat* (°N)	*Lon* (°E)	*dp* (°)	*dm* (°)
CP01	N	0/7	7/7		57.7	32.4	4.4	189.0	57.7	32.4	4.4	189.0		65.1	206.8	4.7	6.5
CP02	N	0/6	6/6		59.7	23.6	2.6	687.6	59.7	23.6	2.6	687.6		72.2	206.2	2.9	3.8
CP03	N	0/6	6/6		57.2	22.3	2.1	987.4	57.2	22.3	2.1	987.4		72.4	215.6	2.3	3.1
CP04	N	0/6	6/6		37.0	46.2	5.7	139.9	62.8	3.4	5.7	139.8		86.6	166.3	7.0	8.9
CP05*	N	0/6	6/6		56.4	63.8	5.9	130.2	29.7	110.6	5.9	130.1	G	41.6	192.8	6.2	8.5
CP06	N	0/7	7/7	H	62.6	14.6	3.8	259.3	62.6	14.6	3.8	259.3		79.2	193.6	4.6	5.9
				L	64.6	138.8	42.4	3.0	64.6	138.8	42.4	3.0		5.8	146.4	54.6	68.0
CP07	N	0/6	6/6		63.4	6.4	3.7	325.4	63.4	6.4	3.7	325.4		84.5	174.9	4.6	5.9
CP08	N	0/7	5/7		28.4	329.2	5.8	177.7	57.8	0.6	5.8	177.7		86.4	292.3	6.2	8.5
CP09	N	0/8	7/8		39.4	337.4	3.6	274.9	58.8	22.7	3.6	275.0		72.7	209.7	4.0	5.4
CP10*	N	0/6	6/6		63.9	359.7	4.0	283.6	75.2	96.7	3.6	348.9	G	86.4	116.4	5.0	6.3
Mean (CP01-04,06-09)			8/10		54.1	9.0	15.9	13.1	60.4	16.2	4.3	167.9		77.9	206.7	5.0	6.5
CX02	N	0/8	8/8		52.0	353.3	1.4	1496	52.0	353.3	1.4	1496		78.7	334.4	1.4	2.1
CX03	N	0/7	7/7		55.9	355.2	2.4	640.9	55.9	355.2	2.4	640.9		83.1	332.6	2.5	3.4
CX04	N	0/8	8/8		58.0	0.6	2.7	416.9	58.0	0.6	2.7	416.9		86.3	291.7	3.0	4.0
CX05	N	0/8	8/8		57.8	357.9	2.4	554.2	57.8	357.9	2.4	554.2		85.9	322.3	2.5	3.5
CX06	N	0/7	7/7		59.2	1.6	2.6	553.7	59.2	1.6	2.6	553.7		87.4	271.8	2.9	3.8
CX07	N	0/7	7/7		61.4	2.6	3.2	358.2	61.4	2.6	3.2	358.2		88.0	200.4	3.8	4.9
Mean (CX02-07)			6/6		57.4	358.3	3.2	451.8	57.4	358.3	3.2	451.8		85.6	316.2	3.4	4.6

Site	Age	polarity ratio R/N	N/N0	Directions: before tilt correction I(°)	D(°)	α_{95}(°)	k	after tilt correction I(°)	D(°)	α_{95}(°)	k	VGPs: *Lat* (°N)	*Lon* (°E)	*dp* (°)	*dm* (°)
CH01	N	10/0	10/10	-39.8	133.8	2.4	412.0	-67.0	148.9	2.5	386.4	-67.3	238.7	3.4	4.1
CMJ01	N	8/0	6/8	-50.2	170.6	3.9	300.7	-50.2	170.6	3.9	300.7	-76.4	155.3	3.5	5.2
CMJ02	N	9/1	7/10	-58.1	159.8	4.1	215.7	-58.1	159.8	4.1	215.7	-74.3	202.9	4.5	6.1
CJ01	N	0/8	7/8	54.8	354.1	2.8	470.2	54.8	354.1	2.8	470.2	81.6	333.5	2.8	3.9
CJ02	N	0/7	7/7	53.6	0.4	2.3	678.5	53.6	0.4	2.3	678.5	81.8	296.1	2.3	3.3
CSZ01	N	0/8	8/8	55.4	7.0	3.3	279.9	55.4	7.0	3.3	279.9	81.6	256.4	3.4	4.7
CSZ02	N	0/7	7/7	57.7	353.1	3.9	240.2	57.7	353.1	3.9	240.2	83.4	353.9	4.2	5.7
CB01	N	0/6	6/6	62.3	75.0	4.2	250.4	62.3	75.0	4.2	250.4	37.1	180.6	5.2	6.6
CB02	N	0/6	6/6	62.5	85.4	2.2	973.5	62.5	85.4	2.2	973.5	30.6	175.9	2.6	3.4
CB03	N	0/6	6/6	65.4	70.0	3.1	480.4	65.4	70.0	3.1	480.4	41.9	177.7	4.0	5.0
CB04	N	0/6	6/6	16.6	130.2	4.4	235.9	65.7	78.0	4.4	235.0	37.1	174.4	5.8	7.1
CB05	N	0/6	6/6	68.1	70.3	2.2	934.1	68.1	70.3	2.2	934.1	42.9	172.8	3.1	3.7
CB06	N	0/6	6/6	68.0	82.5	2.3	828.4	68.0	82.5	2.3	828.4	35.8	169.3	3.3	3.9
Mean (CB01-06)			6/6	60.2	93.0	20.9	11.2	65.4	77.0	3.0	489.2	37.6	175.3	4.0	4.9
CW03	N	0/7	7/7	71.5	343.7	1.7	1214	71.5	343.7	1.7	1214	72.7	87.1	2.7	3.0
CW01	N	10/0	10/10	-54.9	169.2	1.7	842.1	-54.9	169.2	1.7	842.1	-79.2	172.8	1.7	2.4
CW02	N	8/0	7/8	-58.7	169.3	5.4	123.5	-58.7	169.3	5.4	123.5	-81.4	192.3	6.0	8.1

Site	Age	polarity ratio R/N	N/N0	Directions: before tilt correction I(°)	D(°)	α_{95}(°)	k	after tilt correction I(°)	D(°)	α_{95}(°)	k	VGPs: *Lat* (°N)	*Lon* (°E)	*dp* (°)	*dm* (°)
CHD01	N	8/0	8/8	-52.7	172.0	3.1	324.2	-52.7	172.0	3.1	324.2	-79.1	156.5	2.9	4.2
CHD02	N	7/1	6/8	-52.5	173.8	4.2	251.4	-52.5	173.8	4.2	251.4	-79.5	149.1	4.0	5.8
CHD03	N	10/0	10/10	-52.9	170.4	3.2	225.6	-52.9	170.4	3.2	225.6	-78.4	162.3	3.1	4.5
CM01	N	8/0	6/8	-39.1	166.2	1.9	1232.5	-57.4	175.6	1.7	1529.0	-84.7	159.8	1.8	2.5
CM02	N	7/0	5/7	-42.4	141.6	9.8	61.4	-63.4	193.0	5.6	187.4	-80.2	8.4	7.0	8.9
CM03	N	7/0	7/7	-39.6	138.7	6.0	101.3	-66.9	169.2	6.1	98.4	-79.5	257.3	8.4	10.1
CM04	N	7/0	7/7	-49.0	149.8	3.5	295.3	-60.4	202.1	3.5	295.2	-73.5	24.4	4.1	5.4
Mean (CM01-04)			4/4	-43.0	149.1	11.8	61.7	-62.6	185.4	9.2	100.0	-85.6	1.6	11.3	14.4
CS01	N	8/0	8/8	11.5	268.1	1.7	1065.7	-56.3	231.1	1.7	1067.1	-50.8	19.5	1.8	2.4
CS02	N	9/0	9/9	-18.0	259.2	3.8	185.4	-54.3	225.0	3.6	205.8	-54.4	26.0	3.6	5.1
CG03	N	5/0	5/5	-41.8	196.3	8.6	80.8	-41.8	196.3	8.6	80.8	-67.1	77.6	6.4	10.5
		6/0	6/6	-9.9	1.9	8.5	62.8	-9.9	1.9	8.5	62.8	42.4	296.2	4.4	8.6

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Table 2. (*continued*)

Site	Age	polarity ratio R/N	N/N0	Directions								VGPs			
				before tilt correction				after tilt correction				*Lat*	*Lon*	*dp*	*dm*
				I(°)	*D*(°)	α_{95}(°)	*k*	*I*(°)	*D*(°)	α_{95}(°)	*k*	(°N)	(°E)	(°)	(°)
ZG01	N	8/1	6/9	-53.2	137.0	4.4	233.9	-53.2	137.0	4.4	233.9	-55.4	206.7	4.2	6.1
ZY01	N	9/0	8/9	-49.3	184.2	2.5	495.0	-49.3	184.2	2.5	495.0	-78.6	96.2	2.2	3.3
ZY02	N	8/0	8/8	-47.3	186.0	7.1	63.4	-47.3	186.0	7.1	63.4	-76.4	91.9	5.9	9.1
ZL01	N	0/8	8/8	47.6	13.0	2.4	615.5	47.6	13.0	2.4	615.5	73.8	249.7	2.1	3.2
ZL03	N	0/9	8/9	54.2	16.9	7.9	43.5	35.2	23.2	7.9	43.5	60.7	245.3	5.2	9.1
ZL04	N	6/0	6/6	-32.2	158.4	2.1	993.1	-59.6	151.6	2.1	992.6	-68.5	212.6	2.4	3.2
ZL05	N	6/0	6/6	-48.1	90.6	5.8	135.3	-55.4	136.9	5.6	135.3	-56.2	210.4	5.9	8.2
ZL06	N	0/6	6/6	74.2	358.0	3.4	393.9	74.2	358.0	3.4	393.9	70.4	111.8	5.5	6.1
ZL07	N	0/3	3/3	78.7	29.1	12.3	101.8	78.7	29.1	12.3	101.8	58.6	134.9	22.0	23.3
Mean direction from 49 sampled sites															
			49/51					61.5	8.2	4.2		83.4	192.8	A_{95} =6.2	

G: Geographic coordinates, H: High temperature component, L: Lower temperature component

these rocks are very stable, and as the overprinting in DY07 and DZ07 is in the reversed and not the normal direction, we may conclude that these data are acceptable as the middle Jurassic magnetic direction.

3.2 Cretaceous rocks (Zuoyun and Baiqi formations)

In the upper Cretaceous samples, the behaviour of the natural remanent magnetization (NRM) against thermal demagnetization depended on the intensity of the NRM, which ranged from 10^{-3} to 10^{-5} A m^{-1}. In weakly magnetized rocks ($\sim 10^{-5}$ A m^{-1}), the NRM showed two components; the direction of the low-temperature component was erratic (Fig. 3a). The remanence directions became noisy again at temperatures higher than about 600 °C. On the other hand, the strongly magnetized rocks ($\geq 10^{-4}$ A m^{-1}) showed a single and stable component of magnetization. Such contrasting behaviour was observed even among samples obtained at the same site (Fig. 3a, b).

Among the four sites, the uppermost site (DZ01) contained reversely magnetized rocks; among the 12 samples measured, nine were reversed and three normal.

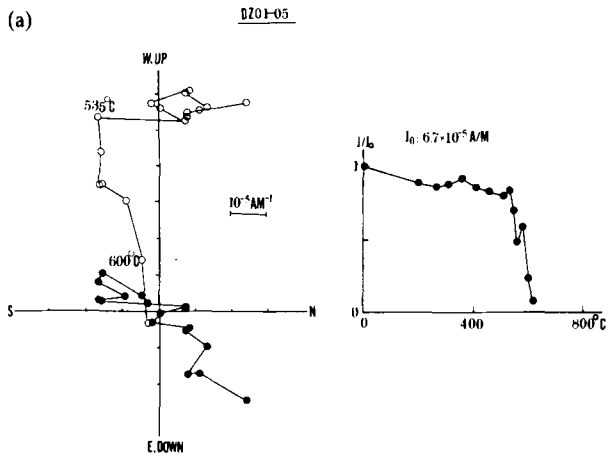


Figure 3. Examples of orthogonal projection of the magnetization vectors in the Cretaceous (and upper Jurassic) rocks from the Datong area. (a) Weakly magnetized DZ01-5 and (b) strongly magnetized DZ01-12 from limestones of Zuoyun formation. (c) DZ03-11 from the same formation shows a very stable remanence carried by haematite.

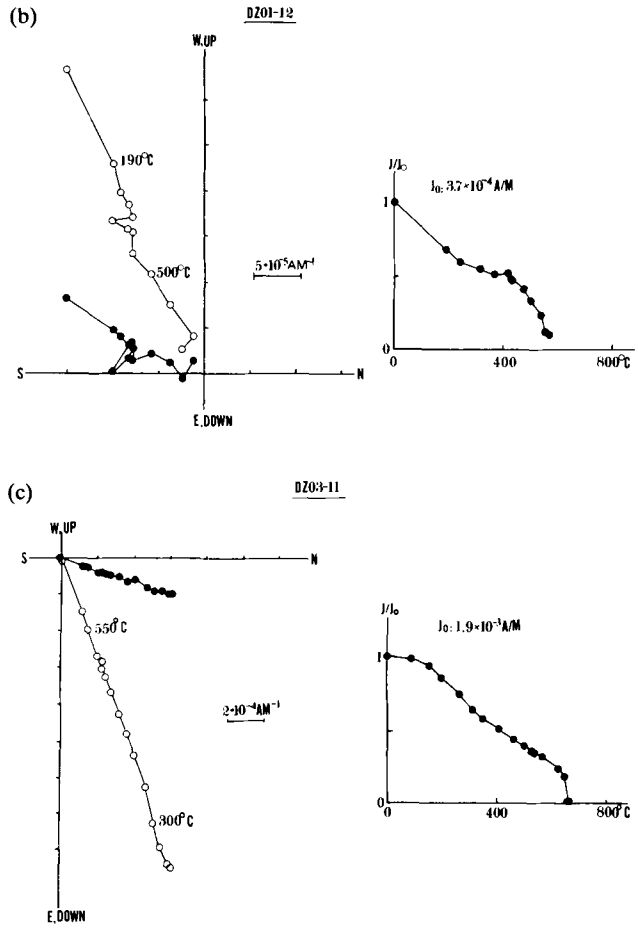


Figure 3. (*continued*)

The two directions are approximately antipodal (Fig. 4). These samples were collected from a layer approximately 0.5 m in height in the thick limestone exposed at the Zuoyun quarry. We conclude from the mixed polarity of DZ01 that both normally and reversely magnetized layers exist in close succession. Three other upper Cretaceous sites showed normal polarity with very stable magnetizations which did not change direction in thermal demagnetization up to 670 °C (Fig. 3c), suggesting that these remanences are carried by haematite.

In the NRM of the Baiqi formation (upper Jurassic to

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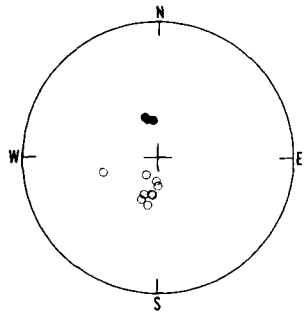


Figure 4. Directions of primary magnetizations in the site DZ01 at the Zuoyung limestone quarry plotted on an equal area projection. Solid and open symbols indicate lower and upper hemispheres, respectively.

lower Cretaceous), a single stable component was observed. AF demagnetization was effective for this formation, but the thermal method was more powerful in demagnetizing all the remanence. The mean direction of magnetization is very steeply inclined compared with all other directions of NRMs from Jurassic and Cretaceous rocks in this study. As this formation was sampled from a single basaltic andesite lava flow, this direction should be regarded as a spot sampling of the secular variation at that age. For this reason, the results from Baiqi formation were excluded from further discussion.

Results of palaeomagnetic measurements are summarized in Table 2. The mean direction for four sites of Zuoyun formation was calculated to be $D = 12.4°$ and $I = 63.4°$ with the 95 per cent confidence limit of $\alpha_{95} = 4.2°$. The cone of 95 per cent confidence of the site DZ02 includes the present axial dipole direction (Fig. 5b). However, in view of the existence of the reversed magnetization and of the very stable behaviour of the remanences in demagnetization experiments, it is concluded that these directions are primary.

3.3 Tertiary rocks (Chifeng and Zhangjiakou)

The Tertiary basalt samples from Chifeng and Zhangjiakou areas contain NRMs which are quite stable and well behaved. Almost all the samples showed a single component of remanence in thermal and AF demagnetizations. Only at one site (CP06), a strong secondary component was detected, and the primary component was reached after the thermal demagnetization step of 350 °C (Fig. 6a). The secondary component was probably acquired by a lightning strike because their directions were scattered and because the samples were obtained from a pillar-like basalt column left over after the quarrying. In all other sites, remanences decayed linearly to the origin in stepwise demagnetizations (Fig. 6b). As the NRMs are very well behaved in demagnetizations, there is no difficulty in choosing primary directions for these rocks. In most of the measurements for basaltic rocks, the sites belonging to the same platform gave very similar results, while different platforms showed different directions. This suggests that, while the centre of volcanic activity migrated with time, each platform was built within a short time before significant secular variation occurred. One of the good examples of such behaviour is the platform CB. In this platform, the site CB04 was from a tilted column and shows a quite different *in situ* NRM direction. After the tilt correction, however, all the mean direction from six sites (CB01–06) are tightly grouped with large easterly declinations (Table 2 and Fig. 8). As there is no field evidence of local rotation due to fault or fold, these means should correspond to the actual field direction. Since the NRM in CH01, which is about 15 km SW from the CB sites, has a reversed polarity, it may be that the CB direction corresponds to a time of polarity transition or excursion. This is only a speculation, but again suggests that the time involved in the formation of a basaltic platform is rather short.

An exception to this conclusion is the platform at Wangyingzi, Chifeng, where the samples were obtained at the top (CW01), the middle (CW02), and the foot (CW03) of the hill, respectively. The remanences from sites CW01 and CW02 show reversed directions, while CW03 is normally magnetized. The two reversed directions of sites CW01 and CW02 are statistically different from each other at the 99 per cent confidence level according to the method of McFadden & Jones (1981). Therefore, all three directions at this platform are independent, which suggests that eruption on this platform occurred in at least three different stages.

Columnar joints are well developed in the Chifeng basaltic platforms. Most of them are vertical, but slanted ones are also observed in some of the platforms. It is an interesting question if the slanting is an original feature or caused by some local tectonic movement. To answer this, detailed sampling was carried out on the Lingshang platform in Pingzhuang, Chifeng. Cores were obtained from five sites with vertical columns (CP01–03, 06, 07), and from five sites with slanted columns (CP04, 08, 10). The result is somewhat confusing. In three slanted columns (CP04, 08, 09), the directions of the *in situ* NRM were quite different from those in vertical columns but became very similar after the direction of column is rotated to the vertical (Fig. 7). On the other hand, in two other slanted columns (CP05, 10), the original *in situ* NRM directions were more similar than the tilt corrected directions. As the remanences behave similarly to thermal demagnetization, it is unlikely that one group retains the original remanence while the other group is remagnetized in a later event. A more plausible explanation may be that originally slanted columns were formed at sites

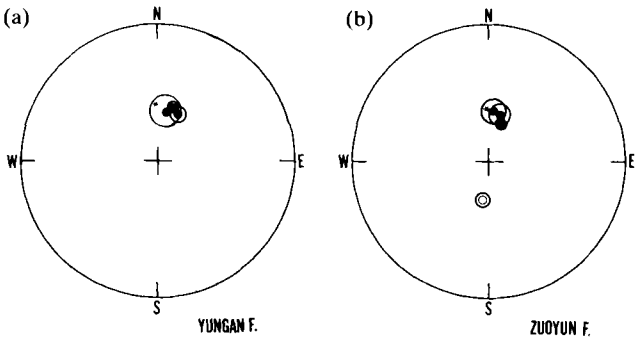


Figure 5. Directions of the site mean primary magnetization with 95 per cent confidence circles in (a) the middle Jurassic Yungan formation and (b) the upper Cretaceous Zuoyun formation. The star indicates the present axial dipole direction. Equal area projection.

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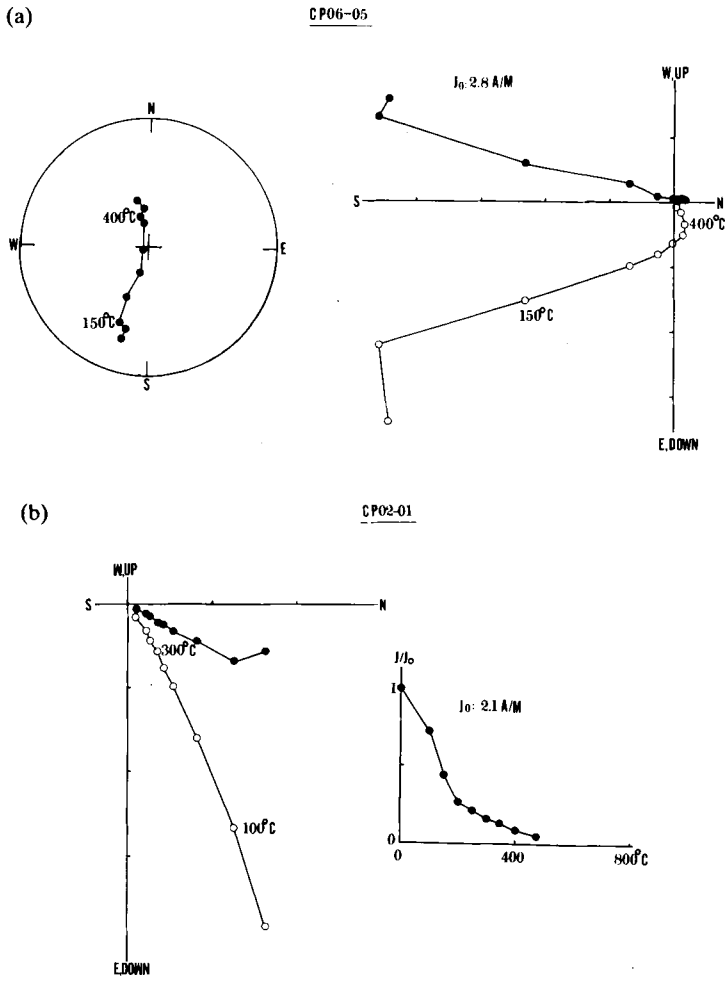


Figure 6. (a) Equal area plot and orthogonal plot of demagnetization results for CP06–5, and (b) orthogonal plots for CP02–1 samples from basalt lavas of Chifeng area. The unit on the vector plot scales is 0.5 A m^{-1}.

CP05 and CP10, because the surface of the lavas was not flat, and because the temperature gradients in these sites were not vertical but slanted. On the other hand, the other group (CP04, 08, 09) was formed with vertical columns but became tilted later by some local movements. This interpretation is partially substantiated by field observations:

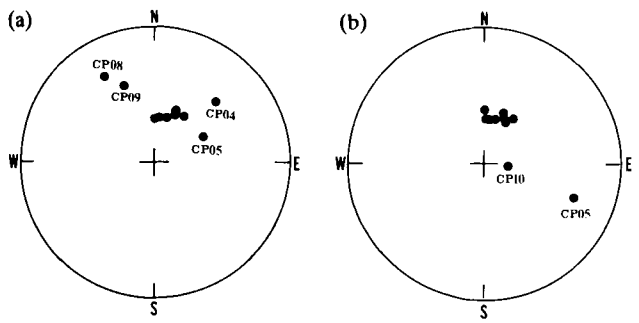


Figure 7. Site mean directions of the basalt platform CP in Pingzhuang, Chifeng (a) before and (b) after tilt correction for the slanting of columnar joints. Dispersion becomes smaller in three slanted columns, but becomes larger in two others. Equal area projection on the lower hemisphere.

at CP08 and CP09 sites, the continuation of the slanted columns is seen to become nearly vertical in the lower horizons. At other three sites, such changes of directions of columns were not observed.

In three other platforms (CB, CM, CS), both vertical and slanted columns were sampled and results are in good agreement when the corrections were applied. We therefore used the tilt corrected data in the following, and discarded the results from CP05 and CP10 which did not give positive fold test. In all, samples were collected from 52 sites (42 in Chifeng, 10 in Zhangjiakou), but the data from 49 sites (40 in Chifeng, 9 in Zhangjiakou) are used below for calculation of the mean direction. One site in Zhangjiakou (ZL02) was very much weathered and a well-defined direction was not obtained by stepwise demagnetization.

Figure 8 shows the mean direction of magnetization for each site. Individual data are summarized in Table 2. Normal and reversed directions were obtained from 29 sites and 20 sites, respectively. The normal and reversed direction are almost antipodal, which confirms that these remanences are stable and reliable. The overall mean direction obtained from 49 site mean directions is $I = 61.5°$, $D = 8.2°$ with $\alpha_{95} = 4.2°$, which is not significantly different from the present axial dipole direction at the site.

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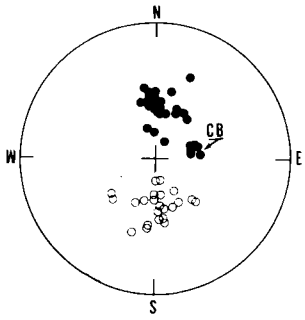


Figure 8. Mean direction of remanences for individual site mean directions in the Chifeng and Zhangjiakou areas. Solid and open circles are on the lower and upper hemispheres of an equal area projection.

4 DISCUSSION AND CONCLUSIONS

Table 3 lists the palaeomagnetic poles obtained in the present study for the NCB. It also includes poles for the Siberia–Europe block, the NCB and SCB for the Jurassic to Tertiary reported from other palaeomagnetic studies. For the SCB, we use the data compiled by Zhai (1988). For the Siberia–Europe block, the compilation by Sasajima & Maenaka (1989) is used; this was obtained by critical assessment of the data listed by Irving & Irving (1982) and by Khramov (1984). Following Zhao *et al.* (1990), we assumed that Europe, Siberia, and Kazakhstan were amalgamated before the Jurassic.

The selection of these reference poles may seem somewhat problematical. Zhai's thesis is written in Chinese and is difficult to obtain for people outside China. However, most palaeomagnetic results reported by Chinese people are written in Chinese, to begin with, and Zhai's thesis has given a detailed critical assessment of the works published by reseachers both in and outside of China, so that it gives an acceptable APWP for SCB at the present stage. For the Siberia–Europe block, Westphal *et al.* (1986) also gave a good data compilation. With their data used in place of the Sasajima & Maenaka compilation, our conclusions are not significantly changed because the two data sets are not very different.

For the Neogene pole for the NCB, we combined our own data with the data reported by Zhao *et al.* (1990), who measured basaltic rocks from 13 sites in Jiening and Chifeng. They also reported results from Cretaceous and Jurassic rocks in Inner Mongolia. These results were obtained after AF or thermal demagnetization. Fold tests or reversal tests were available for some of them.

Fang *et al.* (1988) reported palaeomagnetic results from sandstones of the Yungan formation collected in Jingle area about 200 km south of Datong. Their poles are quite close to the poles we obtained from the same formation in the Datong area. However, we do not include their data in our compilation, because the details of sampling sites, demagnetization methods, or tilt correction are not available. Moreover, we found that all of our samples collected from Cambrian to Jurassic rocks in the Jingle area suffered strong overprinting.

Although the middle Jurasssic and the late Cretaceous mean poles have 95 per cent confidence limits which partially overlap, application of the statistical method developed by McFadden & Jones (1981) shows that these two directions are different at the 90 per cent probability level. Together with the Neogene pole obtained from igneous rocks of Chifeng and Zhangjiakou, we have established an APWP for the NCB since the Jurassic (Fig. 9a).

In deriving the APWP for the NCB, we excluded data obtained from east of the Tanlu fault. They include the

Table 3a. Palaeomagnetic data from the North China Block.

Study area	Site		Age	N	Pole			Refrence
	lat.	long.			lat.	long.	A_{95}	
Chifeng, Inner Mongolia	42.0-42.5	118.6-119.3						
Zhangjiakou,Hebei province	41.0-41.1	114.6-114.7	N	49	83.4	192.8	6.2	This study
Jiening,Chifeng, Inner Mongolia	40.9, 42.0	113.2, 119.2	N	13	88.6	29.2	8.2	Zhao etal,1990
Mean				62	**85.2**	**192.5**	**5.1**	
Datong,Shanxi	40.1	112.9	K_2	4	**79.6**	**170.1**	**5.8**	This study
Datong, Shanxi	40.2	112.8	J_2	3	**76.2**	**199.9**	**8.3**	This study
Jingle, Shanxi	38.5	112.1	J_2	3	76.1	214.0	12.8	Fang etal,1988

Table 3b. Palaeomagnetic pole positions since the Jurassic for the North Eurasia and North and South China Blocks.

Age	Noth Eurasia			Noth China block			South China block		
	lat.	long.	A_{95}	lat.	long.	A_{95}	lat.	long.	$A_{95}(\alpha_{95})$
N_2	85.4	215.1	2.5						
N				85.2	192.5	5.1			
N_1	84.9	192.8	2.9						
E							85.2	173.8	(11.9)
K_2	75.1	163.7	8.3	79.6	170.1	5.8	76.3	172.6	10.3
K_1	72.4	164.9	5.1				73.7	204.3	5.9
J_3	75.0	158.9	9.5				77.0	208.0	11.6
J_2	68.5	142.2	10.0	76.2	199.9	8.3	71.1	203.3	7.3
J_1	74.4	130.5	9.7						
T_{r3}	58.2	137.4	33.7				54.6	202.0	11.7
T_{r2}	50.2	140.7	5.5				54.6	209.1	(5.7)

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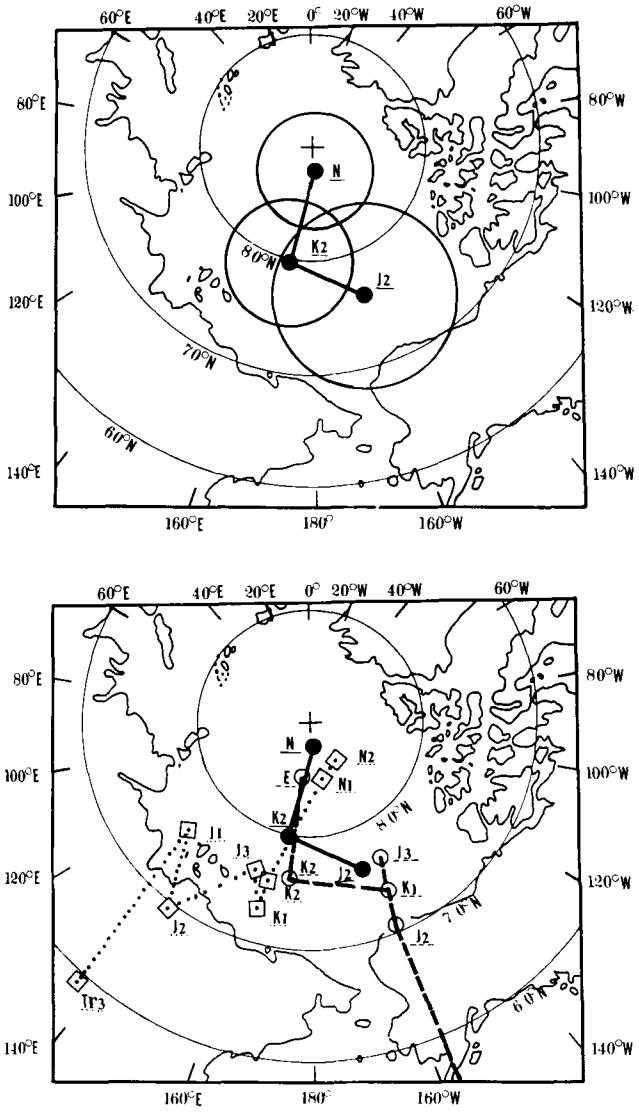


Figure 9. Palaeomagnetic poles for the period of Jurassic to Tertiary (a) with 95 per cent confidence circles for the North China Block, and (b) combined APWPs for the NCB (solid circles), SCB (open circles) and Siberia–Europe Block (squares).

poles reported by Lin (1984) from Shandong Province, and those from the Korean peninsula by Otofuji *et al.* (1983) and Lee *et al.* (1987). They were not used in the present compilation because there appears to be systematic deviation between the poles obtained to the east and west of the Tanlu fault (Uchimura *et al.*, in preparation).

From the comparison of the APWPs for the Siberia–Europe block, the NCB and the SCB, it can be seen that the late Cretaceous poles are very close to each other but the Siberian middle Jurassic pole is displaced from the contemporary Chinese roles (Fig. 9b). Based on the polar wander paths since Carboniferous, Zhao *et al.* (1990) suggested that North China Plate including Outer Mongolia, Inner Mongolia, and the North China block collided with the Siberia–Europe block sometime in the late Permian near the western end of the southern Siberian fold belt. After this initial collision, the North China Plate rotated by about 117° anticlockwise, closing the sea between the two blocks starting from the western end and migrating eastward with time. Zhao *et al.* (1990) concluded that the suturing and rotation were almost completed by the late Jurassic.

Kimura *et al.* (1990) proposed that the two Chinese blocks collided the late Triassic and the suturing continued through the Jurassic until the early Cretaceous. The present data are consistent with this model: the palaeomagnetic poles from the NCB and SCB are quite different in the Palaeozoic (Lin, Fuller & Zhang 1985; Zhao *et al.* 1990) but their displacement decreases with time in the Mesozoic and their poles become very close in the Jurassic. Our data suggests that the two blocks were firmly connected by the middle of the Jurassic. The collision with Siberian block, on the other hand, seems to have taken place a little later. The difference of the pole positions in the Jurassic is still significant, but that in the Cretaceous is not. We can conclude that the amalgamation of the three blocks was almost complete in the Cretaceous. Minor adjustments may have still occurred within the three blocks since then, but the three blocks were essentially within the same tectonic framework since the Cretaceous.

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